TRƯỜNG ĐẠI HỌC QUỐC TẾ MIỀN ĐÔNG
EASTERN INTERNATIONAL UNIVERSITY

BECAMEX BUSINESS SCHOOL

RESEARCH REPORT

TOPIC TITLE:

Factors Influencing Customer Satisfaction and Dissatisfaction with English Learning Apps in Vietnam: A Sentiment and Topic Modeling Analysis of Google Play User Reviews.

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FINAL REPORT

1. Abstract

This study investigates the factors influencing customer satisfaction and dissatisfaction with English learning applications in Vietnam by analyzing the user reviews of ten popular apps from the Google Play Store. Towards the rapid development of mobile learning technologies and the growth of demand for English proficiency in Vietnam. Hence, mobile apps have become common tools for language learning, particularly English. However, user experiences with these apps vary significantly, making it important to understand the key factors that influence user satisfaction.

The main objective of this research is to identify user sentiment and the major topics discussed in online reviews of English learning apps. Moreover, the study aims to analyze how users evaluate different aspects of these applications: learning effectiveness, pronunciation features, user interface design, subscription pricing, and technical performance. The findings highlight several factors that influence user satisfaction and dissatisfaction, providing valuable insights for developers and Educational Technology companies seeking to improve and develop their English learning applications.

Keywords: English Learning Apps, Google Play Store, Sentiment analysis, Deep learning, Edtech.

2. Table of Contents

3. Introduction

3.1 Background

In Vietnam, English proficiency has become an essential skill for academic success and career development. As Vietnam continues to integrate into the global economy, the demand for English communication skills has grown rapidly. Many students, professionals, and self-learners rely on digital tools to improve their English abilities. Among these tools, English learning applications such as Duolingo, Elsa Speak, or Cake have gained widespread popularity. These applications offer interactive learning experiences, including vocabulary exercises, pronunciation training, listening practice, and grammar lessons. The convenience and accessibility of mobile applications make them attractive learning solutions for users with different learning needs.

Despite their popularity, users often report mixed experiences when using these applications. Some users appreciate the effectiveness and convenience of the apps, while others complain about technical problems, subscription costs, or limitations in learning content. Online review platforms such as the Google Play Store provide a valuable source of user feedback. This means users can frequently share their opinions about app performance, features, usability, and overall satisfaction. Besides that, these reviews also contain large amounts of textual data that reflect real user experiences. Therefore, analyzing these reviews can provide important insights into how users perceive English learning applications and what factors influence their satisfaction or dissatisfaction.

3.2 Research Problems

Although English learning applications are widely used, developers often face challenges in understanding user feedback due to the large volume of online reviews. Manually analyzing thousands of user comments is time-consuming and inefficient. To address this challenge, advanced data analysis techniques such as sentiment analysis and topic modeling can be used to automatically analyze large datasets of textual information. These techniques allow researchers to identify emotional patterns and common themes in user feedback. Therefore, this research aims to analyze Google Play user reviews to identify the key factors influencing customer satisfaction and dissatisfaction with English learning applications in Vietnam.

3.3 Research Question

What factors influence user satisfaction and dissatisfaction with English learning applications in Vietnam?

3.4 Research Objectives

- To collect user reviews of English learning apps from the Google Play Store.
- To analyze user sentiment toward these applications using sentiment analysis techniques.
- To identify common themes in user reviews using topic modeling.
- To determine the key factors influencing user satisfaction and dissatisfaction.

4. Literature Review

The rapid development of mobile technology has significantly transformed modern education. Mobile learning, often referred to as M-learning, allows learners to access educational resources anytime and anywhere using smartphones or tablets. This flexibility has made mobile learning increasingly popular worldwide.

Mobile learning (M-learning) has transformed education by enabling flexible and accessible learning through smartphones. In Vietnam, the demand for English proficiency has increased due to globalization, leading to the rapid growth of English learning applications. These apps provide interactive and personalized learning experiences, making them popular among learners (Tran et al., 2025).

However, the large number of available applications creates challenges for users in selecting suitable platforms. Many users rely on online reviews, which significantly influence decision-making and reflect real user experiences. Therefore, analyzing user-generated reviews is essential for understanding customer satisfaction.

Sentiment analysis, a subfield of natural language processing (NLP), is widely used to classify user opinions into positive, negative, or neutral categories. Traditional methods such as Naive Bayes and Support Vector Machines (SVM) have been applied, but they often struggle to capture contextual meaning in text data. Recent studies show that deep learning models, including CNN, RNN, and BERT, provide more accurate results in sentiment classification tasks (Tran et al., 2025).

In evaluating information systems, the DeLone and McLean model is commonly used, emphasizing factors such as system quality, information quality, and user satisfaction. Among these, user satisfaction is a key determinant of system success. Recent research integrates sentiment analysis into this framework by using user reviews to measure satisfaction more effectively.

5. Research Methodology

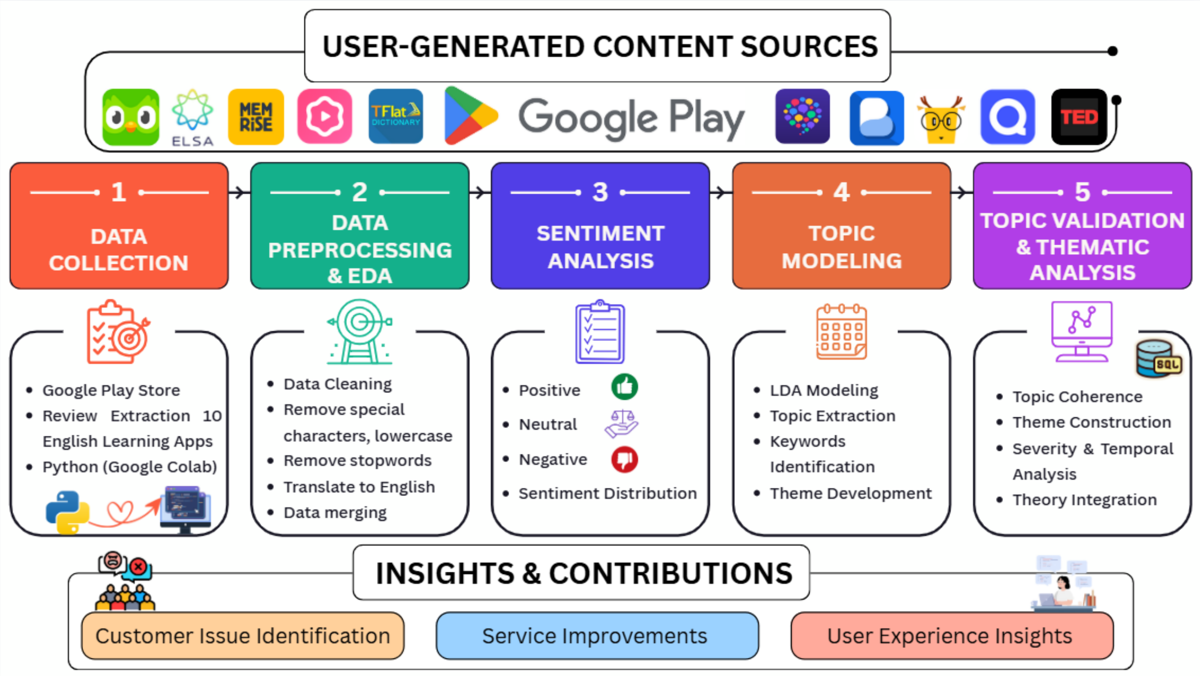

Figure 1: Research Framework for Sentiment Analysis and Topic Modeling of English Learning App Reviews

5.1 Research Design

This study adopts a quantitative research approach using text mining techniques to analyze user-generated reviews from English learning applications.

5.2 Data Collection

User reviews were collected from the Google Play Store for popular English learning apps:

Table 1: The ten most popular apps in the Google Play Store.

App ID	App	Download (Estimates)	Rating	Rows of data
1	Duolingo	500M+	4.7	10000
2	ELSA	10M+	4.5	10000
3	Memrise	10M+	4.7	10000
4	Cake	100M+	4.8	10000
5	TFlat	10M+	4.5	10000
6	HelloTalk	10M+	3.8	1838
7	Busuu	50M+	4.7	1952
8	LingoDeer	10M+	4.6	10000
9	Quizlet	50M+	4.7	10000
10	TED Talks	10M+	3.9	834

The initial dataset consisted of more than 70,000 user reviews collected from the Google Play Store. The majority of the reviews were written in Vietnamese, reflecting the target user base in Vietnam.

The collected dataset consists of several key variables extracted from user reviews on the Google Play Store. The main columns included in the dataset are:

- review_id: A unique identifier assigned to each review
- app: The name of the application (e.g., Duolingo, ELSA Speak, Cake)
- user: The username or identifier of the reviewer
- rating: The numerical rating given by the user (typically on a 1–5 scale)
- comment: The original review text provided by the user
- date: The date when the review was posted

After that, the "user" variable was removed as it was not relevant to the research objectives, which focus on textual analysis and sentiment rather than individual user information.

The application names in the dataset were transformed into numerical identifiers to improve data management and facilitate analysis: Duolingo (1), ELSA (2), Memrise (3), Cake (4), TFlat (5), HelloTalk (6), Busuu (7), LingoDeer (8), Quizlet (9), and TED Talks (10).

5.3 Data Storage

The collected review data was processed and analyzed using Python in Google Colab. The dataset was stored and managed using a PostgreSQL database, which allowed efficient handling and organization of large volumes of textual data. The database was managed through pgAdmin for data storage and querying.

The dataset was organized using a relational database structure in PostgreSQL. Specifically, the data were divided into two main tables following a star schema design:

- A dimension table (list_app), which stores information about the applications, including their identifiers and names
- A fact table (comments), which contains the main review data, such as ratings, review text, and timestamps

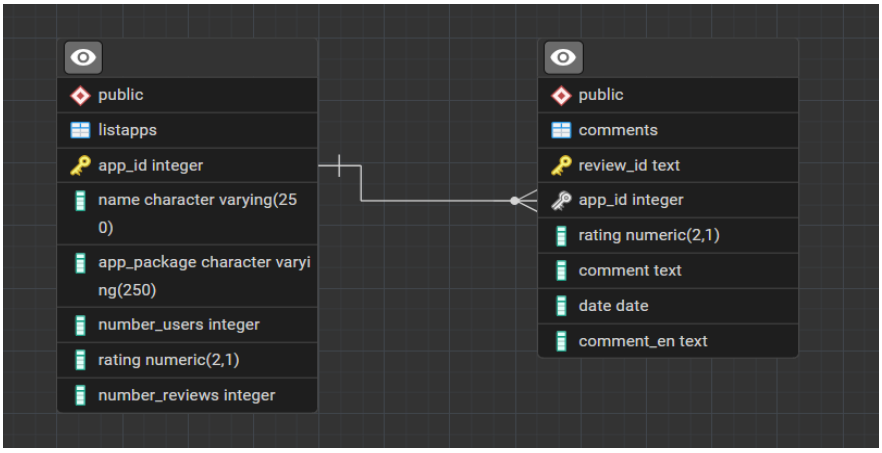

Figure 2: The database schema of listapps and comments table

This structure improves data organization, reduces redundancy, and enables more efficient querying and analysis.

5.4 Data Preprocessing

5.4.1 Data filtering

The initial dataset consisted of 73,436 user reviews collected from the Google Play Store.

To ensure data quality and meaningful analysis, a filtering process was applied. Reviews containing fewer than 25 words were removed, as they often lacked sufficient context for accurate sentiment classification and topic modeling.

After preprocessing, the final dataset used for exploratory data analysis (EDA) consisted of 5,049 reviews.

After the data collection process, the dataset was filtered to remove low-quality reviews, resulting in a final dataset of 5,049 reviews.

5.4.2 Data translation

To facilitate the translation process, the dataset was divided into four subsets based on sentiment categories, including two subsets of positive reviews (Positive 1 and Positive 2), one subset of negative reviews, and one subset of neutral reviews. This division helped manage the translation process more efficiently.

All reviews were then translated into English. The translated text was stored in a new column named comment_en, which was used as the standardized input for subsequent preprocessing and analysis.

The review data underwent several preprocessing steps before analysis to prepare the dataset for sentiment analysis and topic modeling. These steps aimed to improve data quality, reduce noise, and ensure that the text was in a suitable format for natural language processing techniques.

These steps included:

- Removing URLs, numerical values, punctuation, and other irrelevant characters to reduce noise in the text
- Converting all text to lowercase to ensure consistency across the dataset
- Removing standard English stopwords, as well as custom stopwords (e.g., “im”, “dont”, “cant”, “app”), to focus on meaningful content
- Translating non-English reviews into English to maintain a consistent language for analysis
- Tokenizing the text into individual words for further processing
- Handling negation by modifying words following negation terms (e.g., “not_good”) to better capture sentiment context

- Removing rare words that appeared fewer than three times to reduce noise and improve analytical robustness
- Applying part-of-speech tagging and lemmatization using NLTK to convert words into their base forms
- Creating a new column, processed_text, by combining the cleaned and lemmatized tokens into a structured text format for analysis
- The processed_text column was used as the primary input for sentiment analysis, as it represents a cleaned and standardized version of the original reviews.
- Additionally, bigram modeling was applied using Gensim to capture meaningful word pairs, which further supported the topic modeling process.

5.5 Data Analysis Methods

The processed dataset was analyzed using Python in Google Colab. Two main analytical techniques were applied:

- Sentiment Analysis: was used to classify user reviews into three categories: positive, negative, and neutral. This technique helped identify the overall emotional tone of user feedback toward the applications.
- Topic Modeling: Latent Dirichlet Allocation (LDA) topic modeling was applied to identify common themes discussed in user reviews. This technique allowed the study to discover major topics such as learning effectiveness, pronunciation features, user interface design, subscription pricing, and technical issues.

In addition, the cleaned dataset was imported into PostgreSQL using pgAdmin for data storage and management purposes. This approach allows for more efficient handling of large datasets, supports structured querying through SQL, and facilitates easier data retrieval for further analysis and visualization. Moreover, using a relational database ensures better data organization, scalability, and reproducibility of the analysis process.

5.6 Ethical Considerations

Data was collected from publicly available sources from the Google Play Store without personal user information.

5.7 Limitations

Despite the usefulness of the analytical approach adopted in this study, some limitations should be acknowledged, as the dataset consists of voluntary user reviews from the Google Play Store, which may introduce response bias. This means users who post reviews often have particularly positive or negative experiences, meaning the data may not fully represent the opinions of all users of English learning applications.

Another limitation is related to the use of automated sentiment analysis techniques. Sentiment models may have difficulty accurately interpreting informal language, slang, or sarcasm commonly found in online reviews, which may lead to potential misclassification of sentiments.

6. Data Analysis & Findings

6.1 Dataset Overview

Before preprocessing the data for sentiment and topic modeling, the dataset contained 5,048 rows, with 18 missing values in the comments column. After removing the missing values, the dataset consisted of 5,030 rows.

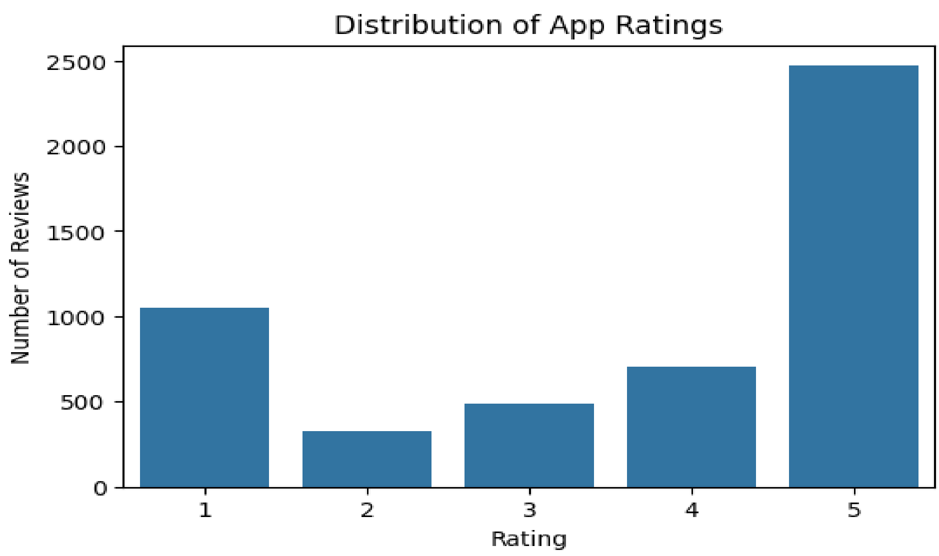

Figure 3: Frequency of User Ratings Across Apps

The chart shows that most comments give a 5-star rating, indicating that these English learning apps are widely trusted and well-received by users. However, the number of 1-star ratings is the second highest, which suggests that there are still significant issues in the user experience. This not only highlights a limitation but also represents an opportunity for the apps to improve and develop further.

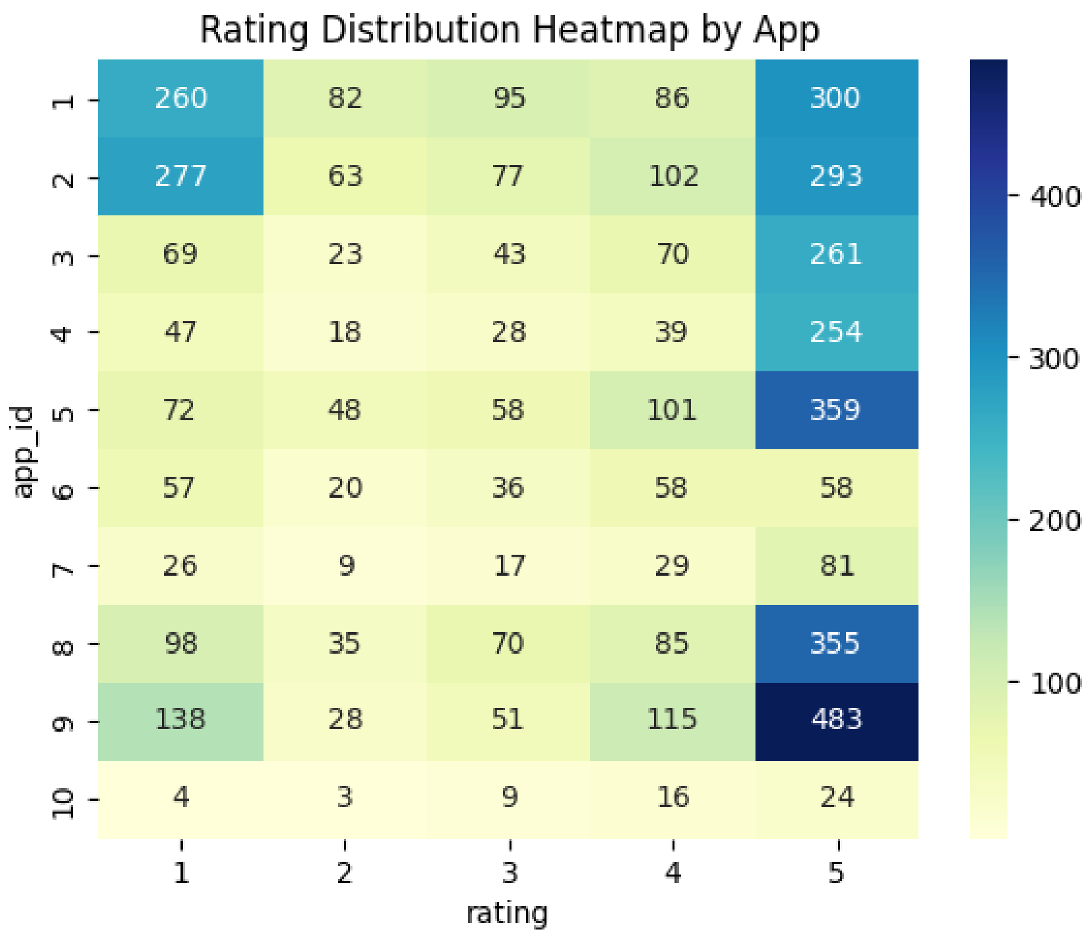

Figure 4: User Rating Patterns Across Apps

The heatmap shows that most apps receive a high number of 5-star ratings, indicating generally strong user satisfaction. Apps like 9, 5, and 8 stand out with the highest positive feedback, suggesting they perform the best among users.

However, some apps, especially Apps 1 and 2, also have many low (1-star) ratings, which points to potential issues or mixed user experiences. Meanwhile, App 10 has very few ratings overall, indicating low user engagement or popularity.

Overall, the data suggests that while most apps are well-rated, user feedback tends to be polarized, with more extreme ratings (high or low) rather than moderate ones.

6.2 Sentiment Analysis Results

The sentiment analysis was conducted on the comments, classifying each comment as positive, negative, or neutral. Out of the total comments, 3,761 were classified as positive, 946 as negative, and 323 as neutral. A new column, sentiment, was added to the dataset to indicate the sentiment classification for each comment.

These results show that the majority of user feedback is positive, suggesting a generally favorable perception of the app. The negative and neutral comments, while smaller in number, provide insights for potential areas of improvement.

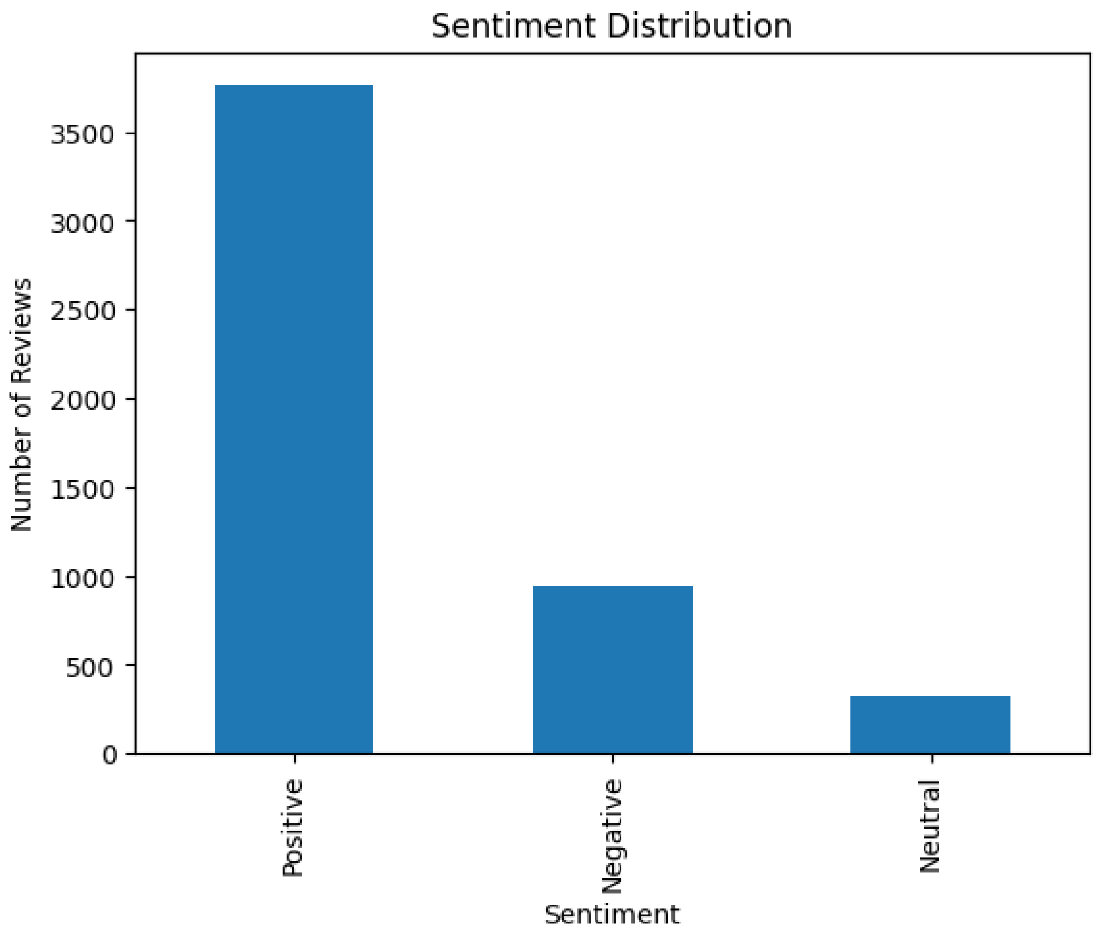

Figure 5: Sentiment Distribution of User Reviews

6.3 Topic Modeling Results

For the group of positive comments, there are three main topics, each characterized by its most frequent words:

Figure 6: Positive topic 0 (use, word, good, learn, feature, like, add, look, application, translation)

Figure 7: Positive topic 1 (learn, good, english, application, help, like, vocabulary, word, language, many)

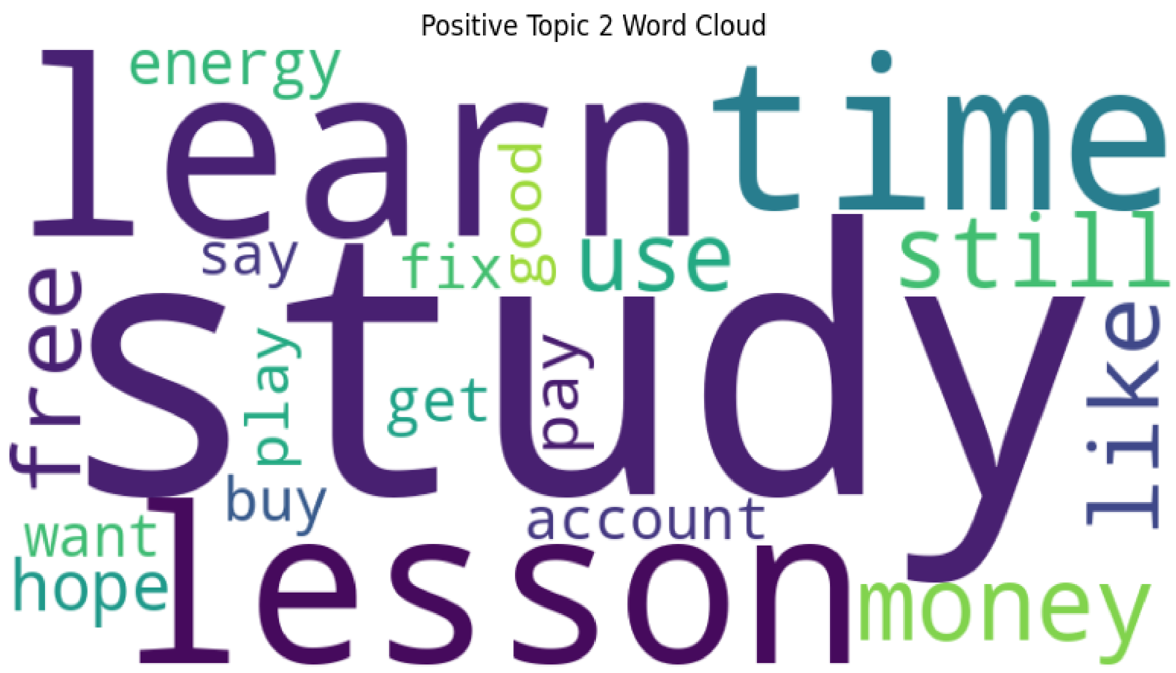

Figure 8: Positive topic 2 (study, learn, lesson, time, money, still, like, free, use, hope)

Similarly, for negative reviews:

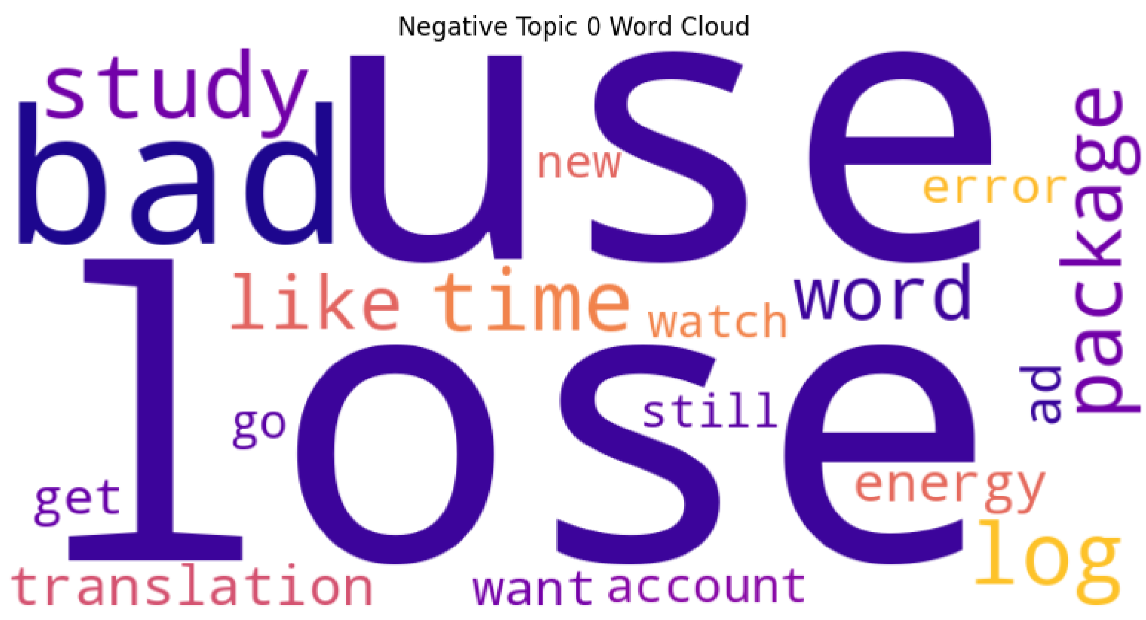

Figure 9: Negative topic 0 (lose, use, bad, study, log, time, package, word, like, translation)

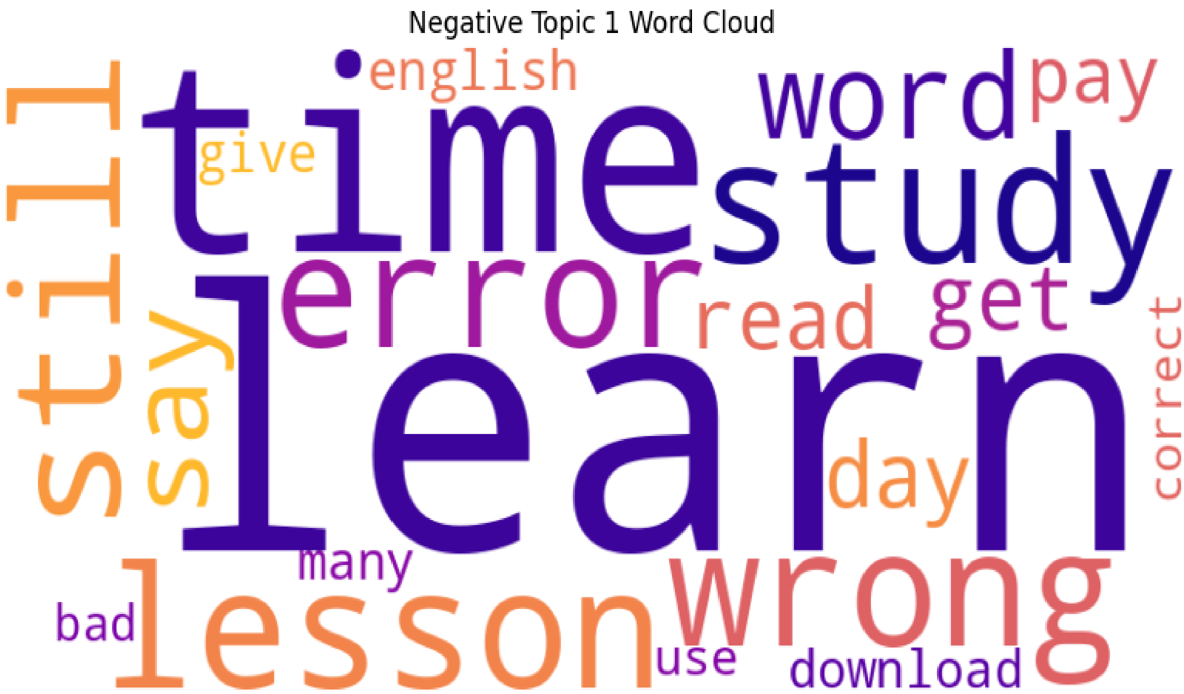

Figure 10: Negative topic 1 (learn, time, study, lesson, wrong, still, error, word, say, day)

Figure 11: Negative topic 2 (money, use, error, still, bad, part, see, fix, word, make)

6.4 Satisfaction Factors

Topic 0: App Features & Usability

- Ease of use
- Useful features
- Translation function
- User-friendly interface
- Continuous improvement (adding features)

Topic 1: Learning Effectiveness (English & Vocabulary)

- English learning support
- Vocabulary improvement
- Helpful for language learning
- Educational value
- Positive learning experience

Topic 2: Cost & Learning Convenience

- Free/affordable usage
- Time flexibility
- Self-study support
- Accessible learning
- Value for money

Positive reviews highlight three main aspects of user satisfaction. First, users appreciate app features and usability, emphasizing ease of use, user-friendly interfaces, useful features such as translation functions, and continuous improvements through new feature updates. Second, many users value the learning effectiveness of the apps, noting that they support English learning, improve vocabulary, and provide meaningful educational experiences. Finally, cost and learning convenience are important factors, as users mention affordable or free access, flexibility in learning anytime, support for self-study, and overall good value for money.

6.5 Dissatisfaction Factors

Topic 0: Technical Issues & System Errors

- Login problems
- Data loss during use
- System instability
- Performance issues
- Disrupted user experience

Topic 1: Content Accuracy & Learning Quality

- Incorrect answers/content
- Translation errors
- Lesson quality issues
- Learning inefficiency
- User confusion

Topic 2: Pricing & Value Concerns

- Expensive packages
- Poor value for money
- Paid features dissatisfaction
- Unresolved errors despite payment
- Low user trust

Negative reviews focus on three main areas of dissatisfaction. First, users frequently report technical issues and system errors, including login problems, data loss, system instability, and poor performance, which disrupt the overall user experience. Second, concerns about content accuracy and learning quality are common, with users mentioning incorrect answers, translation errors, low-quality lessons, and confusion that reduces learning effectiveness. Finally, pricing and value concerns are highlighted, as users feel that some apps are too expensive, offer poor value for money, and include paid features that do not meet expectations, leading to reduced trust, especially when issues remain unresolved after payment.

7. Discussion and Recommendations

This study aimed to identify the key factors influencing user satisfaction and dissatisfaction with English learning applications in Vietnam by applying sentiment analysis and topic modeling to Google Play user reviews. The findings reveal a clear contrast between positive and negative user experiences, providing valuable insights into user expectations and areas for improvement.

The results show that user satisfaction is strongly associated with app usability, learning effectiveness, and cost efficiency. Users tend to have positive experiences when applications are easy to use, provide helpful features such as translation tools, and continuously improve through updates. A user-friendly interface plays a critical role in enhancing engagement, as learners prefer applications that are intuitive and require minimal effort to navigate. This aligns with previous research emphasizing the importance of system quality in determining user satisfaction.

In addition, learning effectiveness is a major driver of positive sentiment. Users value applications that support vocabulary improvement, provide clear explanations, and create meaningful learning experiences. This suggests that content quality and educational value are central to the success of English learning apps. Applications that successfully combine interactive learning with practical outcomes are more likely to retain users and receive positive feedback. Another important factor contributing to satisfaction is cost and convenience. Users appreciate apps that offer free or affordable services, flexible learning schedules, and support for self-study. In the context of Vietnam, where many learners seek cost-effective solutions, pricing plays a significant role in shaping user perceptions. Applications that deliver strong value for money tend to achieve higher satisfaction levels.

On the other hand, user dissatisfaction is mainly driven by technical issues, poor content quality, and pricing concerns. Technical problems such as login failures, system instability, and data loss significantly reduce user trust and negatively impact the overall experience. These issues disrupt the learning process and often lead to frustration, highlighting the importance of maintaining reliable system performance. Furthermore, content-related issues are a major source of negative feedback. Users frequently report incorrect answers, translation errors, and low-quality lessons, which undermine the credibility of the application. Since users rely on these apps for learning, inaccuracies can severely affect learning outcomes and reduce user confidence. This indicates that developers must prioritize content accuracy and quality assurance. Pricing concerns also play a critical role in dissatisfaction. Many users feel that some applications are too expensive or do not provide sufficient value for their cost. Dissatisfaction increases when paid features fail to meet expectations or when technical issues remain unresolved despite payment. This not only affects user satisfaction but also damages long-term trust and brand reputation.

Based on these findings, several recommendations can be proposed. First, developers should focus on improving system performance and stability by minimizing bugs, ensuring smooth operation, and providing reliable data storage. Regular updates and technical support are essential to maintaining user trust. Second, it is important to enhance content quality and learning effectiveness. Developers should ensure that lessons are accurate, well-structured, and aligned with learners' needs. Incorporating feedback mechanisms and expert validation can help improve content reliability. Third, companies should adopt transparent and flexible pricing strategies. Offering free trials, tiered subscription plans, and clear value propositions can help users perceive the application as affordable and worthwhile. Ensuring that paid features deliver real benefits is essential to maintaining user satisfaction. Finally, developers should continue to leverage user feedback through sentiment analysis and data-driven approaches. By continuously monitoring reviews, companies can quickly identify issues, understand user needs, and improve their products accordingly. This approach not only enhances user satisfaction but also supports long-term competitiveness in the EdTech market.

8. Conclusion

This study investigated the factors influencing customer satisfaction and dissatisfaction with English learning applications in Vietnam by analyzing user reviews from the Google Play Store. The main objective was to apply sentiment analysis and topic modeling techniques to identify key themes in user feedback and understand how users evaluate different aspects of these applications. The findings indicate that user satisfaction is primarily driven by usability, learning effectiveness, and cost efficiency. Applications that are easy to use, provide high-quality educational content, and offer good value for money tend to receive more positive feedback. In contrast, dissatisfaction is mainly caused by technical issues, inaccurate content, and pricing concerns. These factors negatively affect user experience and reduce trust in the applications.

The study also provides important implications for developers and EdTech companies. By focusing on improving system performance, ensuring content accuracy, and adopting user-friendly pricing strategies, companies can enhance user satisfaction and increase competitiveness. In addition, the use of sentiment analysis demonstrates the value of data-driven approaches in understanding customer behavior and improving product development. From a business perspective, this research highlights the importance of listening to user feedback and continuously improving digital products based on user needs. In a competitive market, understanding customer satisfaction is essential for retaining users and achieving long-term success.

Finally, this research also provided valuable learning experiences in applying data analysis techniques such as sentiment analysis and topic modeling. It demonstrates how large-scale user-generated data can be transformed into meaningful insights, contributing to both academic research and practical business applications.

9. References

Van, T., Linh, B., & Long, N. (2025). SCIENCE -TECHNOLOGY ANALYZING USER SENTIMENT TO RANK ENGLISH LEARNING APPS ON GOOGLE PLAY. HaUI Journal of Science and Technology, 61(9). https://doi.org/10.57001/huih5804.2025.345

10. Appendices